

Marshall B. Frye
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Research interests: Defect- and interface-mediated phenomena in ferroelectric and emergent functional materials, with emphasis on nanoscale switching behavior, local electrostatic interactions, and structure-property relationships in oxides and two-dimensional material systems. My work combines thin-film synthesis (PLD, MBE), scanning probe and electrical characterization, and spectroscopic analysis to understand and control functional behavior for next-generation electronic and memory devices.

EDUCATION

Georgia Institute of Technology, Ph.D., Materials Science and Engineering **May 2026**
Thesis: “Interfacial Defect Control for Stabilizing and Modulating Ferroelectricity”
Advisor: Prof. Lauren Garten

University of Florida, B.S., Materials Science and Engineering, Minor Business Administration **May 2021**
Summa Cum Laude – Frederick Rhines Scholar

HONORS AND AWARDS

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- **Presidential Fellowship**, Georgia Institute of Technology (2021–2025)
 - **MRS Graduate Student Silver Award**, Materials Research Society (2026)
 - **Russell and Sigurd Varian Award**, AVS (2025)
 - **Thin Film Division Graduate Award**, AVS (2025)
 - **Acta Student Paper Award**, Acta Materialia, Inc. (2025)
 - **Novelis Graduate Scholar**, Georgia Tech MSE (2024)
 - 1st Place - Electronic Materials, GT MRS Poster Session (2026); ACS Sci-Mix Presentation (2026); TMS Poster Award Finalist (2026)

RESEARCH EXPERIENCE

Garten Research Group, Graduate Research Assistant **Aug 2021-May 2026**

- Led multidisciplinary studies of functional thin films and heterostructures, combining deposition, nanoscale characterization, spectroscopy, and modeling to uncover mechanisms governing ferroelectric and interfacial behavior.
- Investigated defect-mediated switching behavior in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2/\text{Al}_2\text{O}_3$ heterostructures, identifying an internal bias field (0.56 MV/cm) and 3x enhancement in coercive field linked to interfacial charge dynamics. Combined electrical characterization, spectroscopy, and phase-field modeling to correlate defect occupancy with ferroelectric behavior.
- Stabilized the metastable polar $P6_3cm$ polymorph of ScFeO_3 through interfacial engineering of iron oxide interlayers, establishing interlayer epitaxy as a route to stabilize multiferroic rare-earth ferrites.
- Demonstrated ferroelectric switching in 2D SnSe thin films grown by pulsed laser deposition, including the first bulk polarization and PUND measurements in this material system. Developed a wafer-scale PLD growth approach for scalable deposition of two dimensional ferroelectric films with monolayer precision.
- Deposited platinum films with near bulk conductivity ($9 \times 10^6 \text{ S/m}$) and record crystallinity (rocking curve FWHM=0.004°) and smoothness ($R_q < 0.5 \text{ nm}$).
- Synthesized and characterized more than 400 thin films spanning oxide, nitride, metallic, and two-dimensional material systems using PLD, MBE, and electron-beam evaporation.
- Secured \$150,000+ in competitive user-facility access through awarded NSF PARADIM and 2DCC-MIP proposals; led MBE deposition runs at PARADIM (Cornell).
- Managed procurement and refurbishment of >\$100,000 of equipment during lab build-out, including PLD chamber.

Nino Research Group, Undergraduate Research Assistant **Aug 2018-May 2021**

- Developed a sulfonation route to increase functional-group density on nanocellulose for opioid sequestration from water; first-author publication in *Cellulose Chem. Technol.*
- Optimized synthesis of reticulated ceramic foams for solar-thermochemical water splitting; co-author publication in *Int. J. Hydrog. Energy*.

TECHNICAL EXPERTISE

- **Thin-film synthesis:** Pulsed laser deposition (PLD), molecular beam epitaxy (MBE), electron-beam evaporation, photolithography (cleanroom), spin-coating; vacuum and laser refurbishment and upkeep.
- **Ferroelectric characterization:** Piezoresponse force microscopy (PFM), Kelvin probe force microscopy, P(E) hysteresis, positive-up-negative-down (PUND), electrical impedance spectroscopy, first-order reversal curve (FORC).
- **Structural and chemical:** X-ray diffraction (high-resolution, in-plane, and grazing-incidence), X-ray reflectivity, atomic force microscopy, scanning electron microscopy/energy dispersive X-ray spectroscopy, X-ray photoelectron spectroscopy, secondary ion mass spectrometry, Raman spectroscopy, UV-Vis spectroscopy, ellipsometry.
- **Modeling and analysis:** Phase-field simulation (FERRET via MOOSE), Landau–Ginzburg–Devonshire theory, Williamson–Hall and Profex analysis, Python for data extraction and analysis.
- **Languages:** Native English; conversational French and Spanish.

TEACHING AND MENTORING

Teaching Assistant, *Georgia Institute of Technology*

Jan 2022–Dec 2024

Optical, and Magnetic Properties of Materials

Principles of Engineering Materials

- MSE 3015: Principles and Applications of Engineering Materials (undergraduate, ~100 students/semester)
- MSE 4750: Electronic, Optical, and Magnetic Properties of Materials.
- Co-developed problem sets and exam answer keys; led weekly office hours and exam review sessions; delivered guest lectures on ferroelectric and dielectric behavior.
- Contributed to and mediated an interactive class project between students and a panel of professors three times.

Garten Research Group, *Undergraduate Research Mentor*

Jan 2022 – May 2026

- Directly mentored four undergraduate researchers on techniques from sample preparation and deposition through characterization, data analysis, and dissemination with an emphasis on independent design of experiments and scientific writing.
- Trained students on PLD operation, AFM/PFM, and XRD analysis; supported applications to graduate school and industry positions.

Graduate Student Advisory Group, *Mentor*

Jun 2022–Aug 2024

- Provided guidance to new graduate students joining Georgia Tech’s Materials Science program.
- Coordinated with faculty, staff, and the incoming students to help the new student orientation run smoothly.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Department of Energy Basic Energy Sciences, *Early Career Network Representative*

Sep 2024–Oct 2025

- Served as the liaison between the 3D Ferroelectric Microelectronics Energy Frontier Research Center and the broader Department of Energy Basic Energy Sciences community.
- Represented the 3D Ferroelectric Microelectronics to share best practices and foster the development of emerging scientific leaders in energy fields.
- Organized webinars and workshops to promote idea exchange and networking across DOE-funded centers.

Materials Research Society, *Treasurer*

Aug 2024–Dec 2025

- Oversaw the development, allocation, and tracking of a budget for chapter operations, events, and initiatives.
- Fundraised more than \$5,000 for the annual poster symposium held for undergraduate and graduate students.
- Planned and executed 6+ events annually to foster collaboration and networking among students and professionals in the field of materials science.

PEER REVIEWED PUBLICATIONS

(13 total peer-reviewed publications, 7 first author (*), 1 senior/last author (†))

- †Lauren M. Garten, Hannah E. Wolfe, Jeremy Knight, Jonathan Chin, Joshua Wahl, **Marshall B. Frye**. “Wafer-Scale Deposition of 2D SnSe with Monolayer-Thickness Precision” ACS Nano. Lett. In peer review
- ***Marshall B. Frye**, John Wellington-Johnson, Lance Fernandes, Prasanna Venkatesan Ravindran, Bogdan Dryzhakov, Mengkun Tian, Asif Khan, Lauren M. Garten. “Tuning the Coercive Field in Ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2\text{-Al}_2\text{O}_3$ Heterostructures via Interfacial Charge Dynamics” Adv. Mater. In peer review

- ***Marshall B. Frye**, Jonathan R. Chin, Nicholas Parker, Matthew Barone, Steven E. Zeltmann, Darrell Schlom, Lauren Garten. “Stabilizing Metastable Rare-Earth Ferrites on (111) Platinum via an Iron Oxide Interlayer” ACS Appl. Electron. Mater., **8**, 2420-2427 (2026)
- ***Marshall B. Frye**, Jonathan R. Chin, Walter J. Smith, Stephen Daniel Funni, Joshua D. Wahl, Anaranya Ghorai, Charles Paillard, Anna M. Österholm, Judy J. Cha, Thomas Beechem, Lauren M. Garten. “Thickness-Modulated Crystal Structure and Band Gap of 2D SnSe Deposited by Molecular Beam Epitaxy” NPJ 2D Mater Appl., **10** (2026)
- ***Marshall B. Frye**, Mengkun Tian, Eoin Moynihan, Ana Sanchez, Lauren Garten. "Interlayer-Mediated Stabilization of Metastable $P6_3cm$ ScFeO₃ on Al₂O₃" Adv. Mater. Interfaces, (2025)
- ***Marshall B. Frye**, Jonathan Chin, Matthew Barone, Steven Zeltmann, Lauren Garten. “Enhancing the Conductivity and Crystallinity of (111) Platinum Films via a Two-Temperature Deposition and Substrate Annealing” Scr. Mater., **256**, 116442 (2025)
- ***Marshall B. Frye**, Lauren Garten. “Reaching the Potential of Ferroelectric Photovoltaics” Acc. Mater. Res., **4**, (2023)
- ***Marshall B. Frye**, Shangradhanva Vasisth, Amalie Atassi, David Mazyck, Juan C. Nino. “Effect of Degree of Sulfonation in Nanocellulose/Chitosan Composite on Adsorption of Cationic Dye as Opioid Simulant” Cellul. Chem. Technol., **55**, 87-99 (2021)
- Kayla Y. Chuong, Yang Liu, Thomas Forbes, **Marshall B. Frye**, Lukas Graber, Lauren M. Garten. “Determining the Effect of Bi₂O₃ on the Microstructure and Electrical Response of Cold Sintered ZnO Varistors” J. Am. Ceram. Soc., (2026)
- Ian C. Graham, Kayla Chuong, **Marshall B. Frye**, Anna M. Österholm, Lauren M. Garten. “Epitaxial Stabilization and Tunable Oxygen Stoichiometry of R₃₂ BaNiO_{3-x} Thin Films via Pulsed Laser Deposition” J. Mater. Res., **41**, 208-219 (2026)
- Jonathan Chin, **Marshall B. Frye**, Derick Liu, Maria Hilse, Jeffrey Shallenberger, Ke Wang, Roman Engel-Herbert, Mengyi Wang, Yun Kyung Shin, Nadire Nayir, Adri van Duin, Lauren Garten. “Self-limiting stoichiometry in SnSe thin films” Nanoscale, **15**, 9973-9984 (2023)
- Jonathan Chin, Bonnie Gardner, **Marshall B. Frye**, Derick Liu, Sebastian Marini, Jeffrey Shallenberger, Matthew McDowell, Maria Hilse, Stephanie Law, Lauren Garten. “Determining the oxidation stability of SnSe under atmospheric exposure” MRS Comm., **14**, 1000–1006 (2024)
- Elizabeth Gager, **Marshall Frye**, Dylan McCord, Jonathan Scheffe, Juan C. Nino. et al., “Reticulated Porous Lanthanum Strontium Manganite Structures for Solar Thermochemical Hydrogen Production” Int. J. Hydrog. Energy, **47**, 31152-31164 (2022)

SELECTED PRESENTATIONS

- “Ferroelectricity in 2D SnSe Deposited via Pulsed Laser Deposition” MRS Spring Symposium (2026)
- “Two-Step Deposition of Platinum for Heteroepitaxial Growth” TMS 2026, Poster Award Finalist (2026)
- “Thickness Scaling of (2h00) Oriented SnSe Deposited by Pulsed Laser Deposition” ACS Sci Mix (2026)
- “Interlayer-Modulated Coercive Field in Hf_{0.5}Zr_{0.5}O₂ Ferroelectric Devices” AVS 71 (2025)
- “Controlled Growth of (200) SnSe by Pulsed Laser Deposition” Electronic Materials and Applications (2025)
- “Growth of SnSe Thin Films for Bulk Photovoltaic Effect Studies” MRS Fall Meeting (2024)
- “Stabilization of $P6_3cm$ ScFeO₃ on (111) Pt Mediated via an Fe₃O₄ Interlayer” AVS 70 (2024)
- “Growth of Hexagonal ScFeO₃ for Bulk Photovoltaic Effect Optoelectronics” Electronic Materials Conference (2024)
- “Stabilizing Polar Polymorphs of Scandium Ferrite for Photovoltaics” US-Japan Seminar on Dielectric and Piezoelectric Ceramics (2022)
- “Developing Ferroelectric Rare-Earth Ferrites for Solar Energy Harvesting” IEEE Winter FerroSchool Symposium (2022)
- “Functionalized Cellulose-Chitosan Membranes for Adsorption of Amine Compounds” MRS Spring Symposium (2021)

PROFESSIONAL AFFILIATIONS

- Professional societies: Materials Research Society (MRS), American Vacuum Society (AVS), The Minerals, Metals, & Materials Society (TMS).
- Honor societies: Sigma Xi (full member), Alpha Sigma Mu, Tau Beta Pi

REFERENCES

Available upon request